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A Comparative Analysis of Diffusion and Variational Autoencoder Models for Text-Conditioned Facial Reconstruction

Project Proposal

Applied Machine Learning

WBAI065-05.2025-2026

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1 Preliminary domain knowledge

Time can be essential in pursuing criminals. Our project aims to help law enforcers in generating an image of a potential criminal's face, starting from descriptions collected from witnesses. However, beyond this specific forensic application, we also emphasize the broader usefulness of such generative models in general face synthesis and human image generation tasks, including applications in accessibility tools, and human-computer interaction. Our project will be focused on the field of generative image modelling: a subfield of machine learning that focuses on creating new images from either text or another image.

In their paper, Rombach et al. (2022) present latent diffusion models (LDM) as high-performing models for multiple tasks, including text-to-image synthesis. Diffusion models learn from distributions of data by denoising a normally distributed variable. They are a sequence of equally weighted denoising autoencoders. The key idea behind LDMs is to reduce computational demands, while maintaining the same state-of-the-art performance. This is done by introducing the latent space, because it enables the model to extract the semantic information in the data, and train in a space with lower dimensionality, which requires less computation.

We aim to fine-tune a pre-trained text-to-image diffusion model and compare its performance with a Variational Autoencoder (VAE). We want to focus our research around prompt-adherence and image quality. As such, we introduce the following research question: 'How do diffusion-based generative models compare to VAE-based models in terms of image quality and text-prompt alignment on a constrained domain (human faces)?'.

Our hypothesis is that a diffusion-based generative model will outperform a VAE-based model in terms of both image quality and semantic alignment. However, we expect that the diffusion-based generative model will be more computationally expensive, while the VAE-based model will prove to be more interpretable.

2 Preliminary data exploration

For the training and evaluation process, we will use 100,000 randomly chosen samples from the OpenFace-CQUPT dataset with human captions. It consists of around 10 million samples (specifically 10388531) of images found on the internet, with human captions explaining them. The features included in the dataset are: image ID, image name, human caption, image URL, laion caption, and SHA-256 hash signature.

While trying to visually inspect the dataset, we observed that a subset of the images cannot be loaded from the provided URLs (resulting in the data points with missing values). Out of 1500 data points, which we decided to sample for the data exploration, 545 corresponded to invalid or inaccessible images. If we assume that the images that are unavailable are evenly distributed throughout the dataset, then approximately 64% of the dataset has valid images.

Additionally, some images contain multiple individuals or depict nudity. Such cases introduce ambiguity and potential risks for model training, as they may lead to unclear labels or unintended biases. They were not removed from the dataset during the data exploration, but we acknowledge the possible

impact they may have on the model's performance.

Dataset review by Dai et al. (2024)

Research by Dai et al. (2024) evaluated this dataset along four different dimensions: image quality, text quality, image-text alignment, and dataset scale/diversity. Image quality was assessed using metrics such as BRISQUE and CLIPQA. The results show that FaceCaption-15M achieves higher overall image quality compared to several existing facial datasets. Captions were evaluated to be with high linguistic diversity. Image-text alignment was measured using an image-text matching (ITM) score. The dataset demonstrated a strong alignment.

Captions analysis

For the analysis of the captions, we preprocessed them to standardize the textual input. All text was converted to lowercase, non-alphabetical characters were removed, and captions were tokenized.

We analysed the distribution of caption lengths (in number of words) to provide insight into the variability and descriptive richness of the dataset. The histogram (Figure 1) shows that the captions in the dataset consist of between 10 and 260 words each. We can also observe that the majority of the samples have around 10 to 100 words.

While exploring the data, we noticed that a large number of captions start with "In this image/picture" or similar text, which would suggest that the first few tokens of each caption provide no information useful to the model. For this reason, a keyword frequency analysis was conducted to better understand the dominant semantic content. The results from this analysis can be seen in Figure 2. The most frequent words are common conjunctions and determinators. Among most common meaningful and significant for our task words, there are "hair", "image", "wearing", "lipstick", "youthful", "eyes". This analysis will help identify potential biases in the results, such as overrepresentation of certain descriptors.

Furthermore, we defined a set of manually selected attributes related to: gender (e.g. *male*, *female*), age (e.g. *young*, *old*), ethnicity (e.g. *white*, *black*, *asian*), or sensitive content (*nudity*). Each caption was scanned for these keywords. Two analyses were performed. Firstly, attribute frequency distribution, which shows (Figure 3) how often each attribute appears. From the analysis of the 1500 samples we can see that the dataset includes slightly more captions describing women than captions describing men. Moreover, there is a big difference between the number of captions describing young and old people, suggesting final model might have a big difference in performance between those groups. The same counts for the ethnicity attributes. While "black" and "white" are used with almost the same rate, "asian" was part of very few captions, which might effect the performance of the model.

Secondly, attribute co-occurrence matrix (Figure 4) captured how often attributes appear together. In the figure, we can see the percentage of the captions where certain attributes co-occur. This allows us to see correlations between the attributes and find patterns in the data that might lower the performance. Looking at Figure 4 we can see that in some captions words that refer to a woman and words that refer to a man are both used in almost 9% of the captions. Another interesting observation is that young women are described in almost twice as many captions as young men. This suggests that young men are underrepresented as opposed to young women which might affect the model's

predictions. Moreover, we can see that in most cases of nudity the caption refers to a female person and the age of the person is not mentioned.

Finally, to explore the semantic structure of captions, we encoded them using CLIP ViT-B/32 and reduced the resulting embeddings to two dimensions using Principal Component Analysis (PCA). Due to the model constraints, captions were truncated to fit within CLIP's token limit (77 tokens). The resulting 2D visualization is presented in Figure 5. In the figure we can see two regions that have more closely clustered suggesting there might be two descriptions of people/clothes that are more often represented. We can also see that the right half of the plot contains a few scattered data points which might signify outliers or underrepresented groups of people or styles (maybe some types of clothes are more represented than others).

Images analysis

To better understand the visual properties of the dataset, we analysed the distribution of image resolutions for the images we could load (955 in total). We extracted the most common resolutions and visualized their frequency distribution (Figure 6). The analysis shows that the dataset is not uniform in image dimensions, with a small number of resolutions dominating the collection. If we assume that the resolution is evenly distributed throughout the dataset, then we can conclude that around 5.2% of the images that can be loaded have 360×480 resolution, around 2.1% have 180×320 , but overall the dataset includes images with many different resolutions.

Additionally, we decided to further examine the pixel-level characteristics of the images by analysing intensity distributions. We converted 20 sample images to grayscale to inspect overall luminance patterns, and histograms are used to visualize pixel value distributions. The three figures below show the grayscale histograms for three distinct images in the dataset. What we can see is that there are instances of images with different illumination: low (Figure 7 shows that the image has a lot of low frequency signals, signifying darker tones), evenly distributed (Figure 8 shows that the image has a lot of different frequency signals, signifying both darker and lighter tones), and high (Figure 9 shows that the image has a lot of high frequency signals, signifying lighter tones).

2.1 Text-Image Analysis

To assess the correlation between the raw captions and images, we computed the CLIP scores for each data point. With those scores we try to measure the semantic relation between each pair of caption and text. Higher scores mean better correlation between the two components. The distribution of CLIP scores in the 955 samples with available images is shown in Figure 10. As we can see from the figure, most of the caption-image pairs received a CLIP score between 20 and 40. Considering the CLIP scores can vary between 0 and 100, the raw data in this dataset seems to have rather low CLIP scores. However, this is expected in large-scale web-scraped datasets, where captions are often noisy, overly generic, or only weakly descriptive of the image content. For this reason, lower CLIP scores are not necessarily problematic at this stage, as one of the goals of preprocessing is to filter and improve image-text alignment.

3 Proposed preprocessing

Based on the performed data analysis, we decided to implement the following preprocessing steps aimed at improving data quality, reducing noise, and mitigating potential biases.

3.1 Data filtering and cleaning

The methods we chose as part of our pre-processing are based on the results of the data exploration. Figure 11 shows a simple diagram of the preprocessing explained in this section.

During preliminary inspection, we observed that a subset of images could not be retrieved from their URLs. All such instances of missing values will be removed entirely from the dataset to prevent training instability. In addition, we will introduce filtering criteria based on caption content. Samples referring to nudity or multiple individuals will be identified using keywords in the captions and removed, since they may introduce ambiguity and lead to worse quality of the outcomes. Furthermore, images with extremely low resolution will be excluded from the dataset, as they do not contain sufficient visual detail for reliable feature extraction and may negatively impact the quality of the generative model. To further improve data quality and ensure sample uniqueness, we will remove duplicate entries from the dataset. Finally, we will remove samples that exhibit very low CLIP similarity scores between image and caption embeddings. This is motivated by the need to improve image-text alignment quality and reduce noisy signals.

Additionally, after applying all filtering steps, the remaining dataset will be split into training, validation, and test sets using an 80/10/10 split. The training set will be used to optimize model parameters, the validation set will be used for hyperparameter tuning and early stopping, and the test set will be used exclusively for final evaluation of model performance.

3.2 Caption preprocessing

Textual data will be standardized to reduce variability and improve consistency. We are planning to apply the following transformations: conversion to lowercase, removal of non-alphabetical characters, tokenization into word-level units, and removing the first 3–5 tokens of the captions (to remove the repetition of "in this image."/"the picture shows"). To handle the variability in caption length, we will also truncate overly long captions.

3.3 Image preprocessing

As a last step in our preprocessing we will transform the images. In the previous section we mentioned that since we are working with real photos, the brightness of the images is very different. Thus, we will normalize the brightness of the images to ensure consistent quality of the generated images. Additionally, all images will be cropped to focus on the detected facial region. This ensures that the model is trained specifically on facial features rather than background information, improving both relevance and consistency of the generated outputs.

4 Proposed model(s) and baseline

Main model

The model we want to fine-tune is the `Würstchen` model. (Pernias et al. 2023). We chose this diffusion-based text-to-image generation model because it aligns perfectly with the task we want to solve (text-to-image), and, most importantly, it is less computationally expensive than other state-of-the-art models, while still maintaining high quality of images.

We will fine-tune the `Würstchen` model for text-conditional latent image generation, on the image caption and the image itself from the samples we selected from the OpenFace-CQUPT dataset. Whether or not we choose full fine-tuning or Low Rank Adaption (LoRA) is yet to be decided. We will perform hyperparameter tuning on learning rate, batch size, and other parameters, as we see fit.

Baseline model

Our baseline model is a Conditional Variational Autoencoder (CVAE) model (D’Cunha 2025), trained on the text embeddings of the image caption produced by Contrastive Language–Image Pre-training (CLIP) encoder. We will have to experiment with the embedding size and the number of hidden layers of the encoder and the decoder, as well as batch size and other hyperparameters.

Training strategy

Both models will be trained using supervised learning on image-caption pairs. The CVAE is trained using reconstruction loss in combination with a latent regularisation term, while the diffusion model is trained using a denoising objective in latent space. We will use the AdamW optimizer for both models and train using mini-batch gradient descent. Early stopping will be applied based on validation loss to prevent overfitting.

Hyperparameter tuning strategy

We will apply a structured hyperparameter tuning. The following hyperparameters will be tuned:

- learning rate
- batch size
- latent dimensionality (CVAE)
- number of hidden layers (CVAE)
- guidance strength (diffusion model)

We will select the final configuration based on validation performance measured using CLIPScore and FID.

5 Proposed evaluation

5.1 Quantitative Metrics

To assess whether our models are actually successful, we will look at several quantitative metrics that measure quality and accuracy. First, we will use the Fréchet Inception Distance (FID) score (Parmar, Zhang, and Zhu 2022). The idea is to measure how similar our generated faces are to a set of real human faces. If our fine-tuned Diffusion model (Rombach et al. 2022) has a lower FID score than the VAE (Kingma and Welling 2013), it will indicate that it produces more realistic features. Secondly, we will calculate a CLIPscore. This measures the mathematical "alignment" between the prompt and the generated image. This is important because it tells us if the model is actually listening to the descriptions provided by the prompt.

5.2 Evaluation Beyond Performance

We also believe it is important to evaluate the models beyond just their raw performance:

- **Model Complexity vs. Quality:** We plan to compare the complexity of our two models by looking at their total trainable parameters. We want to see if the fine-tuned Stable Diffusion is actually efficient or if it just produces better images because it is much larger than the VAE. This quality-per-parameter assessment will allow us to discuss the practical trade-offs between a fast, simple baseline and a more complex model.
- **Inference Cost:** We will also try to measure the inference cost of both models specifically within the Habroc cluster environment. By comparing the inference times, we can determine the "quality-per-second" value of the Diffusion model, this can help us in justifying its higher resource usage over the lighter VAE baseline.
- **Explainability (XAI) Heatmaps:** To make the model more transparent, we will use heatmaps (Selvaraju et al. 2017) to show which specific parts of the face, such as the eyes or nose, were most influenced by key descriptors in the prompt. This acts as a confidence check to ensure the model is focusing on relevant features identified in our keyword frequency analysis, such as "hair," "lipstick," or "youthful".
- **Bias audit:** We will also conduct a bias audit based on our preliminary findings that identified a significant demographic imbalance in the dataset, including a female-skew and a near-absence of Asian individuals. We need to evaluate how these skews affect generative performance to ensure the suspect images are fair and accurate for all demographic groups.
- **Human Assesment:** We hope to also include human assessment of the models output based on the Labelbox methodology (Maniar 2024). Labelbox used human raters to assess 4 different metrics; Alignment with prompt, Photorealism, Detail and Artifacts. By placing the images side by side with the prompts that generated it, participants can give a score (low, mid, high) on how well it performs on each metric. We will try to create our own version that will help determine whether the generated output is going be useful or not.

6 Model usage

The idea is to use this model to help criminal investigators. The use case would be to generate a visualization of a suspect based on a witness report. The optimal way to make this model efficient for

this purpose is to make sure that it can be used on site. By eliminating the time that a forensic sketch artist has to draw the suspect first, it will likely increase the probability of the suspect being caught. This model could be declared useful if it is as accurate as the forensic sketch artist. Additionally if the model can generate varieties of faces using the same report, this could be useful as well, as it takes more time to draw more sketches.

Before: Witness report needs a forensic sketch artist to generate visuals of the suspect.

After: Witness report is enough to generate the visuals of the suspect.

7 Risk assessment

The most likely obstacle in our project development is time. We chose to work on an ambitious project, which not only requires understanding more complex concepts, but we also chose to fine-tune a diffusion model and train a VAE, which will take time. This will leave us with very small room for errors. In order to avoid this problem, we discussed the timeline of our project thoroughly. We also aim to complete more task simultaneously (writing the report while working on the code).

One possible risk of deploying our model might be bias. If the data is biased, then the predictions might be too. Since we want our model to be used by law enforcers, bias could lead to serious consequences, like arresting innocent people. We are trying to clean the dataset, in order to eliminate biases towards a certain gender, or age, by trying to make the dataset as balanced as possible. Interacting with the data in this manner however, might introduce our own subconscious biases, however.

8 Individual learning outcomes

Batbayar

While the learning goals stated by the course are still top priority for me. I would also like to learn how to work with a group on a larger project using version control properly, as I still have little experience and knowledge about this.

Julia

Beyond the course learning outcomes, I aim to improve my ability to work effectively in a team on larger projects. This includes better planning, task division, and communication within a group setting. I also want to gain more experience with professional collaboration on GitHub, especially in managing larger codebases and using version control workflows more efficiently.

Sophia

I am really excited to be pushing my limits with this project: I have not worked with diffusion models before, and until now have only used toy-datasets. I am also looking forward to working in a larger group than I am used to, and to learn how code management works on GitHub in this, more complex, setting.

Teodora

I think this project is a great way to broaden my knowledge about machine learning. It will be my first time working with diffusion models and extensive analysis and augmentation of images, so I am motivated. I am also looking forward to working in a larger group and learn about team work for bigger projects and efficient ways to collaborate with your teammates.

9 Relation to grading specifications

The project complies with the general requirements, as we use a real-world dataset consisting of images gathered from various sources on the internet. The dataset contains high-dimensional visual and textual data and exhibits class imbalance and inherent biases, which requires a comprehensive preprocessing pipeline. Our final model, Stable Diffusion, will be compared against a baseline Variational Autoencoder (VAE).

We will perform advanced preprocessing based on detailed dataset analysis. The preprocessing steps are designed to ensure that the dataset is suitable for training large-scale generative models and to improve image–text alignment quality. We will perform systematic hyperparameter tuning for both the VAE and the Stable Diffusion model.

We are planning to conduct a comprehensive evaluation of model performance, including both quantitative and qualitative analysis. Quantitative evaluation will rely on metrics such as FID and CLIP-Score, while qualitative evaluation will include visual inspection of generated samples and human assessment of realism and prompt adherence.

The project is novel since it explicitly compares diffusion models and VAEs for the task of face generation conditioned on textual descriptions, which is highly relevant for forensic reconstruction applications. Additionally, instead of relying solely on heuristic cleaning, we incorporate CLIP-based similarity scoring to filter low-quality or misaligned image–text pairs. Furthermore, we perform structured dataset analysis focusing on semantic attributes and bias patterns, which is rarely integrated into standard generative modelling pipelines.

Planned improvements

Transfer learning

A pre-trained Stable Diffusion model will be fine-tuned on the domain-specific dataset, which will improve convergence and enable higher image fidelity.

Advanced evaluation framework

In addition to FID and CLIPScore, we plan to include qualitative human evaluation of generated faces, as well as analysis of prompt sensitivity and robustness under variations in textual input.

Dimensionality reduction and interpretability

We apply PCA on CLIP embeddings to visualize semantic clustering of captions and analyze the latent structure of the dataset. This provides additional interpretability of the learned representations.

Bias and fairness analysis

We explicitly study how demographic attributes present in the dataset influence model outputs. This includes examining how attributes such as age, gender, and ethnicity are distributed and co-occur, and how these patterns may affect generated images. This helps identify potential biases in the model and improves awareness of ethical implications in deployment.

Robustness analysis

We evaluate model behavior under noisy or ambiguous captions, low-quality images, and weak image-text alignment cases. This strengthens the real-world applicability of the model.

Overall, the project fully satisfies all minimum grading requirements. In addition, it introduces several extensions that significantly enhance the scientific depth, novelty, and practical relevance of the work.

10 Timeline

The project is structured over an eight-week period. The overall planning is presented in Table 1. In the first phase (Weeks 1–2), the focus is on writing the project proposal, establishing the research direction, and defining the methodology. The second phase (Weeks 3–4) involves data preprocessing and initial model development, including training a VAE and conducting a literature review to support the introduction section of the final report. In Weeks 4–5, we will fine-tune the Stable Diffusion model and add the model description to the report. The final phase (Weeks 6–7) includes model deployment and analysis, potentially incorporating human evaluation. During this period, the report is also finalized. The project concludes in Week 7 with the final presentation.

11 Member contributions

The member contributions to the proposal are as follows: Batbayar 25%, Julia 25% Sophia 25%, Teodora 25%.

References

- D’Cunha, R. (Jan. 2025). *Conditional Variational Autoencoders for Text to Image Generation*. URL: <https://towardsdatascience.com/conditional-variational-autoencoders-for-text-to-image-generation-1996da9cefcfb/>.
- Dai, Dawei et al. (2024). “FaceCaption-15M: A Large Multimodal Facial Image-Text Dataset”. In: *arXiv preprint arXiv:2407.08515*. URL: <https://arxiv.org/abs/2407.08515>.
- Kingma, Diederik P and Max Welling (2013). “Auto-encoding variational bayes”. In: *arXiv preprint arXiv:1312.6114*.

- Maniar, Krish (2024). *A Comprehensive Approach to Evaluating Text-to-Image Models*. Accessed: 2026-04-30. URL: <https://labelbox.com/guides/a-comprehensive-approach-to-evaluating-text-to-image-models/>.
- Parmar, Gaurav, Richard Zhang, and Jun-Yan Zhu (2022). “On Aliasing and Accuracy in FID Implementation”. In: *Computer Vision and Pattern Recognition (CVPR)*. URL: <https://arxiv.org/abs/2104.11222>.
- Pernias, Pablo et al. (2023). *Wuerstchen: An Efficient Architecture for Large-Scale Text-to-Image Diffusion Models*. arXiv: 2306.00637 [cs.CV]. URL: <https://arxiv.org/abs/2306.00637>.
- Rombach, Robin et al. (2022). “High-resolution image synthesis with latent diffusion models”. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp. 10684–10695.
- Selvaraju, Ramprasaath R et al. (2017). “Grad-cam: Visual explanations from deep networks via gradient-based localization”. In: *Proceedings of the IEEE international conference on computer vision (ICCV)*, pp. 618–626.

Appendices

A Dataset exploration

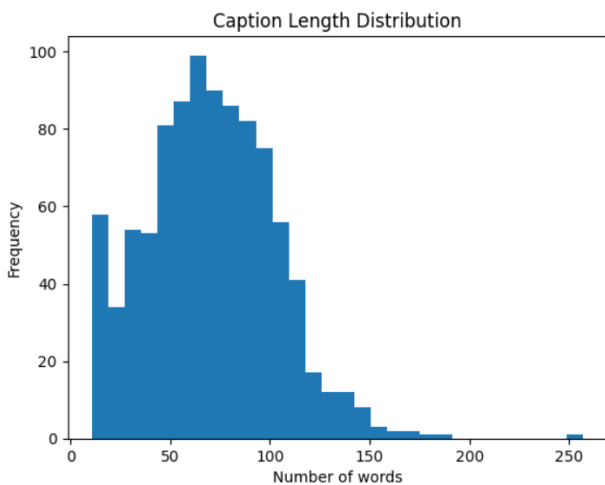


Figure 1: Distribution of caption lengths (number of words) in the cleaned dataset.

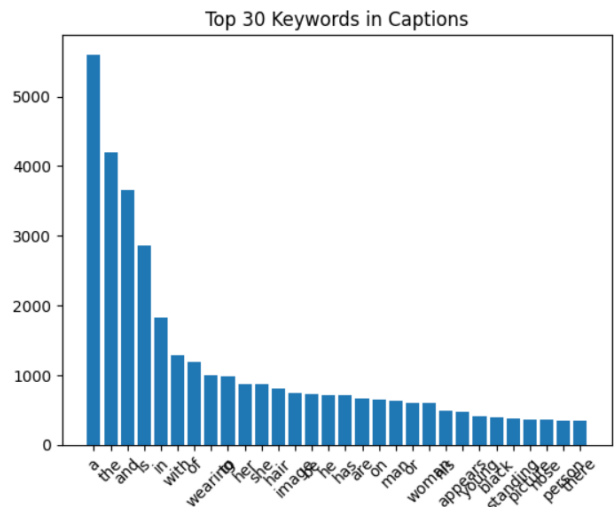


Figure 2: Frequency distribution of the most common words in image captions.

B Semantic attribute analysis

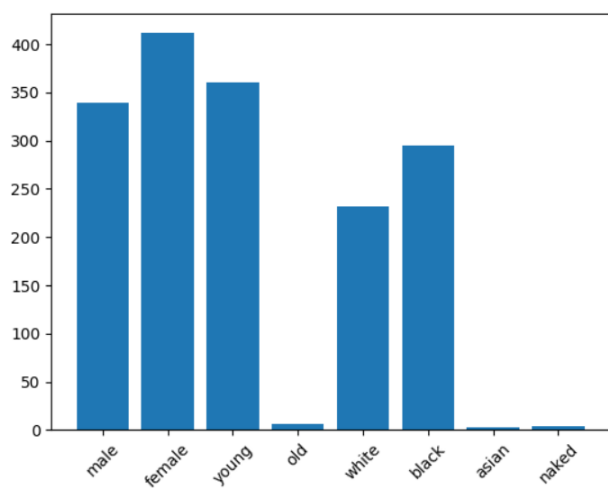


Figure 3: Overall frequency of selected semantic attributes across the dataset.

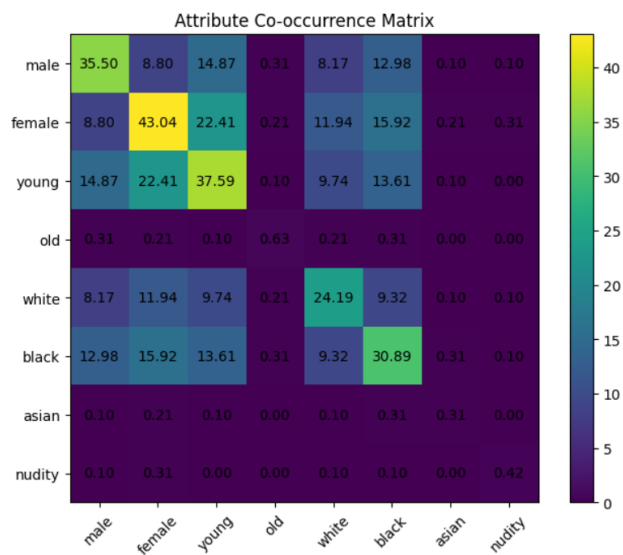


Figure 4: Co-occurrence matrix of extracted semantic attributes from captions.

C Embedding visualization

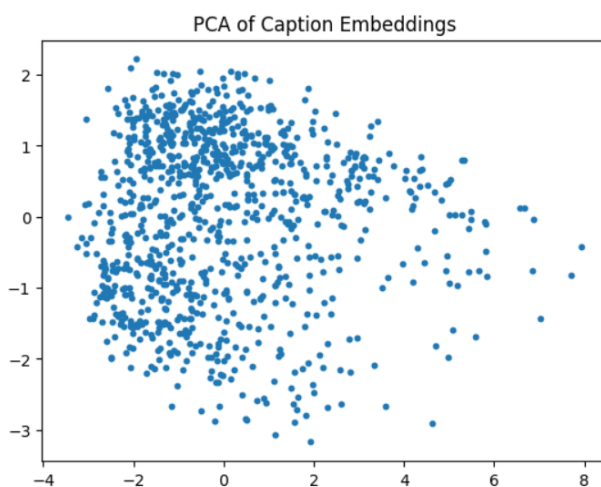


Figure 5: Two-dimensional PCA projection of CLIP text embeddings.

D Image resolution analysis

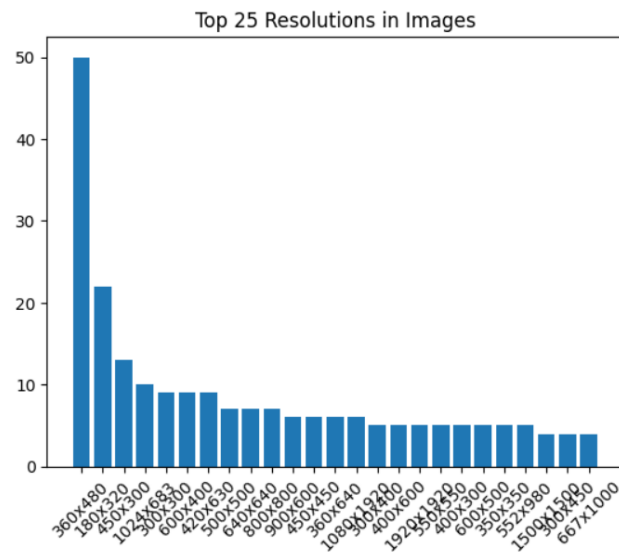


Figure 6: Frequency of the most common image resolutions in the dataset.

E Pixel intensity distributions

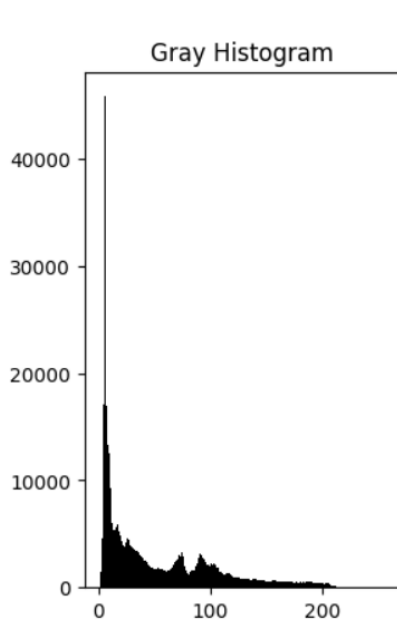


Figure 7: Low brightness sample.

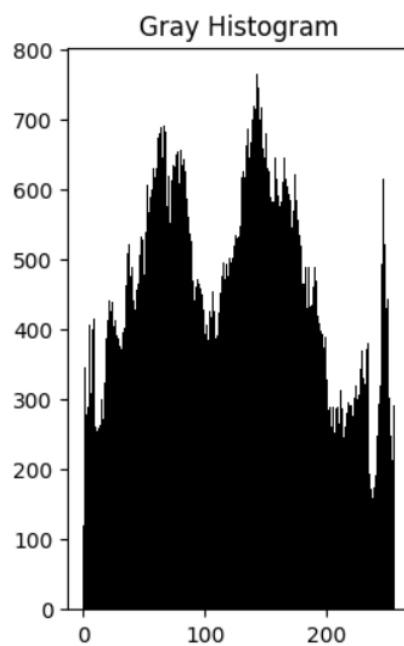


Figure 8: Even brightness sample.

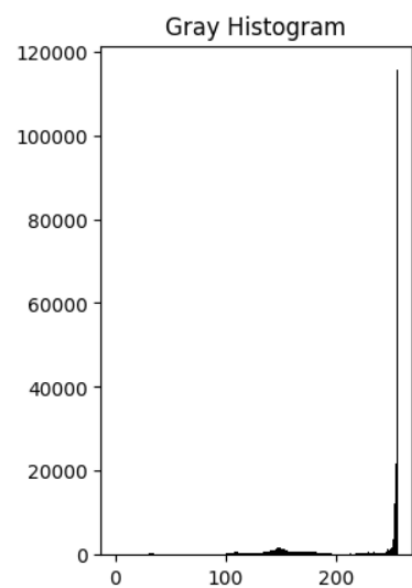


Figure 9: High brightness sample.

F CLIP analysis and preprocessing

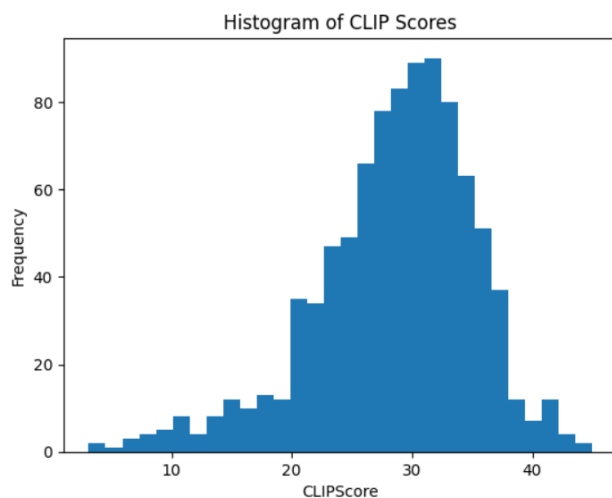


Figure 10: Histogram of CLIP scores.

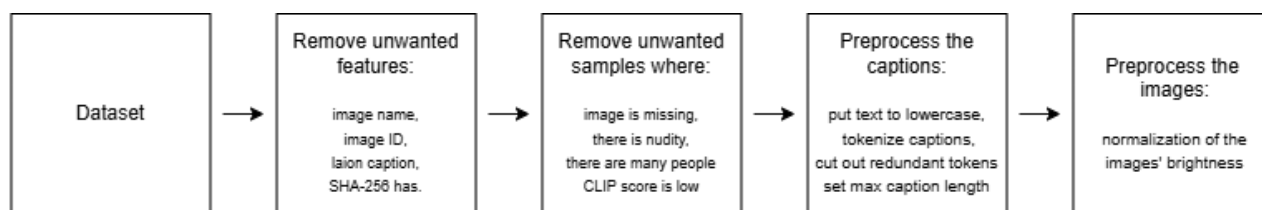


Figure 11: Preprocessing pipeline.

G Project timeline

Table 1: Project timeline over 8 weeks

Task	Week 1 (20–26.04)	Week 2 (27.04–3.05)	Week 3 (4–10.05)	Week 4 (11–17.05)	Week 5 (18–24.05)	Week 6 (25–31.05)	Week 7 (1–7.06)	Week 8 (8–14.06)
Write the proposal								
Data preprocessing								
Training VAE								
Introduction + literature search								
Fine-tuning Stable Diffusion								
Model description								
Model deployment								
Analysis								
Finishing report								
Final presentation								